



Rossmoyne Senior High School

Semester One Examination, 2023

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

If required by your examination administrator, please
place your student identification label in this box

WA student number: In figures

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In words

Circle your Teacher's Name:

Mrs Alvaro	Mr Bartoshefski	Mr Buckland	Mr Liu
Mr Lockyer	Mr Luzuk	Mr Mohammadi	
Ms Narendranathan	Mr Ng		
Mr Stephens	Mr Tanday		

Time allowed for this section

Reading time before commencing work:

five minutes

Working time:

fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	53	35
Section Two: Calculator-assumed	12	12	100	101	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	7	
2	8	
3	8	
4	8	
5	8	
6	7	
7	7	
S1 Total	53	
S1 Wt ($\times 0.6731$)	35%	
S2 Wt	65%	
Total	100%	

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Section One: Calculator-free**35% (53 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

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Question 1**(7 marks)**

(a) Let $f(x) = 2x^2 - 2x + 1$

(i) Determine the equation of the line of symmetry for the graph of $f(x) = 2x^2 - 2x + 1$.

(2 marks)

(ii) Determine the discriminant of the quadratic $2x^2 - 2x + 1 = 0$ and use it to explain the number of real root(s).

(2 marks)

(b) Solve the following equations.

(i) $(2x - 1)(x + 7) = 0$.

(1 mark)

(ii) $x^2 + 2x - 15 = 0$.

(2 marks)

Question 2

(8 marks)

- (a) Determine all solutions of the equation $2 \cos(x - 40^\circ) + \sqrt{3} = 0$ between 0° and 360° .
(3 marks)

- (b) (i) Using $\cos 2x = \cos(x + x)$, show that $\cos 2x = 2 \cos^2 x - 1$.
(2 marks)

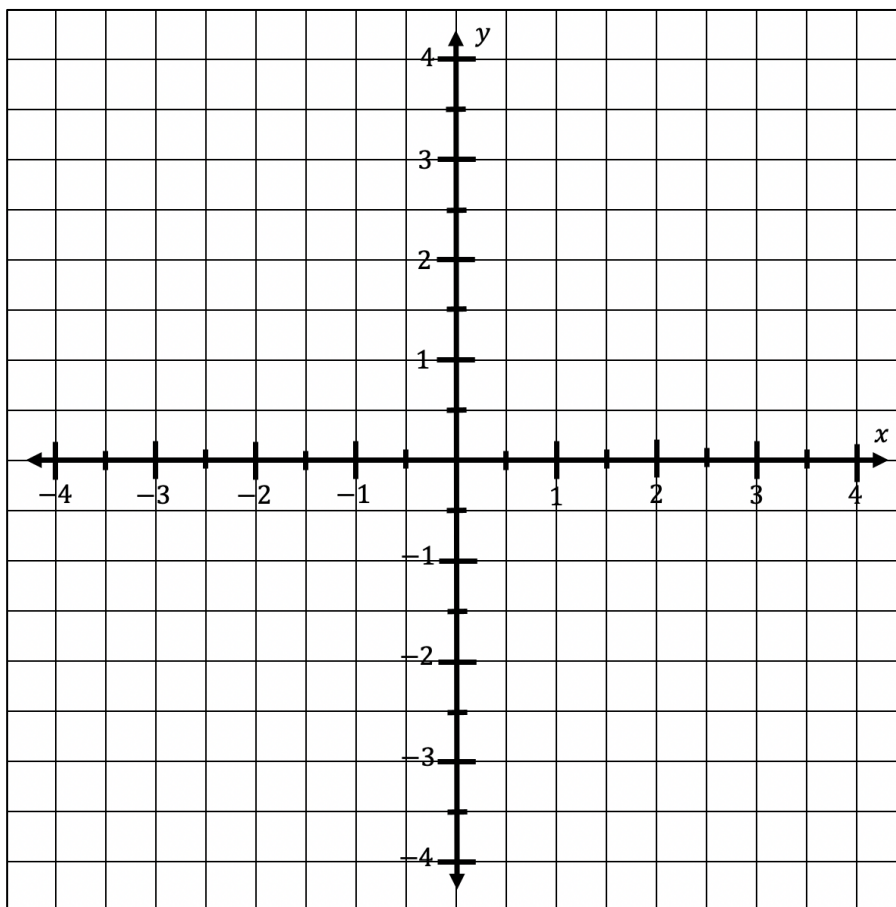
- (ii) Use this identity to obtain an exact value for $\cos 15^\circ$.
(3 marks)

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Question 3

(8 marks)

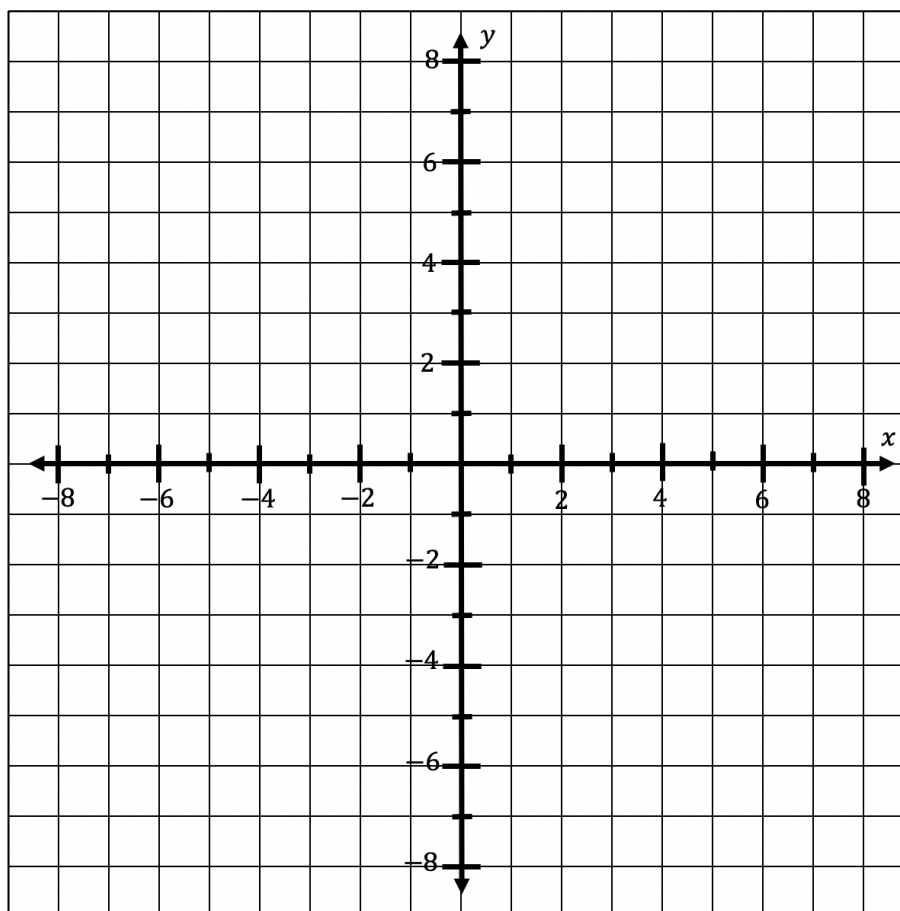
- (a) Sketch the graph of the relationship $x = y^2$ on the axes below, and then use your graph to explain why y is not a function of x . (3 marks)



(b) Another relation has equation $(x - 1)^2 + (y + 4)^2 = 9$.

(i) Sketch the relation on the axes below.

(2 marks)



(ii) Determine the y -coordinate(s) of the point(s) where the line $x = 2$ cuts the graph of the relation. (3 marks)

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Question 4

(8 marks)

(a) Let $f(x) = (x + 2)(8x^2 - 22x + 15)$.

(i) Express $f(x)$ in expanded form, without brackets.

(2 marks)

The graph of $y = f(x)$ has roots $x = -2$, $x = \frac{5}{4}$ and $x = \frac{3}{2}$.

(ii) Express $f(x)$ in the form $(x + 2)(ax + b)(cx + d)$, where a, b, c and d are integer constants. (2 marks)

(b) The graph of $y = 3(x - p)^3 + q$ has a vertical inflection point at $(-2, 24)$ and y -axis intercept at $(0, k)$, where p, q and k are constants.

(i) the value of k .

(2 marks)

(ii) the y -intercept of the graph.

(2 marks)

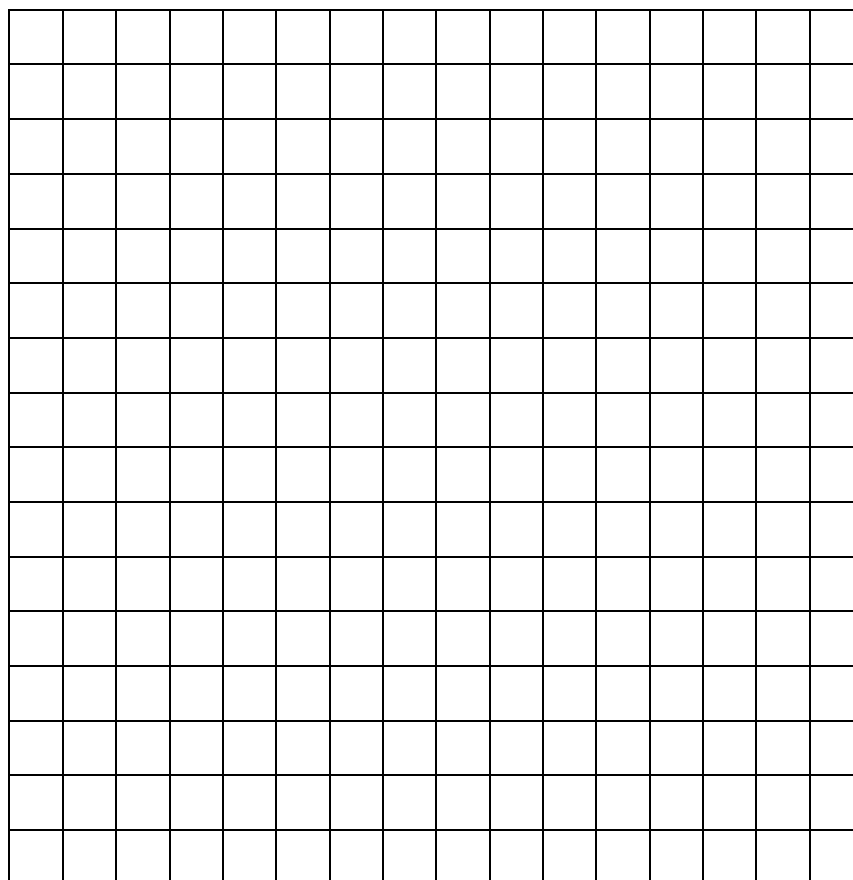
Question 5

(8 marks)

- (a) State the exact value of $\tan\left(\frac{\pi}{6}\right)$. (1 mark)
- (b) Express in simplest form the angle
- (i) 16° in radians. (1 mark)
- (ii) $\frac{3\pi}{10}$ in degrees. (1 mark)
- (c) Given that angle $\theta \in \left\{0^\circ, \frac{7\pi}{8}, 270^\circ, \pi, -100^\circ, \frac{\pi}{2}, 21^\circ, -\frac{\pi}{20}\right\}$, state which, if any, of the angles are such that
- (i) $\cos \theta < 0$. (1 mark)
- (ii) $\tan \theta$ is undefined. (1 mark)
- (iii) $\tan \theta < 0$ and $\sin \theta > 0$. (1 mark)
- (d) Given that $\sin \alpha = \frac{3}{5}$ and $\cos \beta = \frac{1}{5}$, determine the value of $\sin(-\alpha) + \cos(\beta + \pi)$. (2 marks)

Question 6**(7 marks)**

- (a) Sketch the graph of $y = 3(x - 2)^2 - 27$ below.

(4 marks)

- (b) The graph of $y = ax^2 + bx + c$ has roots $x = -10$ and $x = 5$ and passes through the point $(-5, 20)$. Determine the value of a , the value of b and the value of c . **(3 marks)**

Question 7

(7 marks)

Consider the following expansion of the binomial $(x + 1)^{11}$:

$$x^{11} + 11x^{10} + kx^9 + 165x^8 + 330x^7 + 462x^6 + 462x^5 + 330x^4 + 165x^3 + 55x^2 + 11x + 1.$$

- (a) State the degree of the above polynomial and the value of the constant k . (2 marks)

- (b) There are m ways to choose 3 cats and n ways to choose 8 cats from a choice of 11 different cats. Use coefficients from the expansion to explain whether m is greater than, equal to, or less than n . (2 marks)

The first three terms in the expansion of $(x + 2)^{10}$ are $x^{10} + 20x^9 + ax^8$.

- (c) Determine the value of a and the coefficient of the x^7 term in the expansion. (3 marks)

Supplementary page

Question number: _____

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